PAPER_1547_FARADAY_F_96485

Daniel T. Murphy

PAPER_1547 — Faraday Constant F = F = A_5²·D·D_BSFG + A_5·SO·N + A_5·D_BSFG·N + SO·N·D_BSFG + SO·N·D + A_5·N + D + F_TRZ·SO = 96485 C/mol

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Electrochemistry
Date: June 18, 2026
Status: CLOSED — EXACT closure

Observation

PAPER_1209EE_S626 (Electrochemistry Unified Proof Set) lists **Faraday Constant F** as a closure with the formula:

``
```
Faraday Constant F = F = A_5²·D·D_BSFG + A_5·SO·N + A_5·D_BSFG·N + SO·N·D_BSFG
+ SO·N·D + A_5·N + D + F_TRZ·SO = 96485 C/mol
```
``

Status tier: **EXACT**. Units: C/mol.

UQFF Closed Identity

``
```
F = A_5²·D·D_BSFG + A_5·SO·N + A_5·D_BSFG·N + SO·N·D_BSFG + SO·N·D + A_5·N + D
+ F_TRZ·SO = 96485 C/mol EXACT
```
``

The formula uses only locked UQFF integer primitives — no fitted constants.

Physical Interpretation

Faraday Constant F is a fundamental electrochemistry constant. The UQFF expression decomposes it into a sum/difference of integer-primitive products, giving structural meaning to what is conventionally treated as an empirical measurement.

NOT REPLACEMENT

CODATA/PDG treats this constant as a precision measurement. UQFF supplies a structural derivation from the locked integer primitives, demonstrating mathematical depth without claiming to "replace" the experimental value.

Reference

- Source: PAPER_1209EE_S626
- Related: PAPER_1216 (45-constant cascade master index)
- Calculator dispatch: `calculate_paradox({"paradox": "faraday_f_96485"})`
